AI-compute supply chain

Chart pack based on analysis of data from UN Comtrade + TDM + Trade Atlas, from which we construct a bilateral monthly panel of HS6 trade flows capturing the global AI-compute supply chain (60 HS6 codes, 2017-01 to 2026-04).

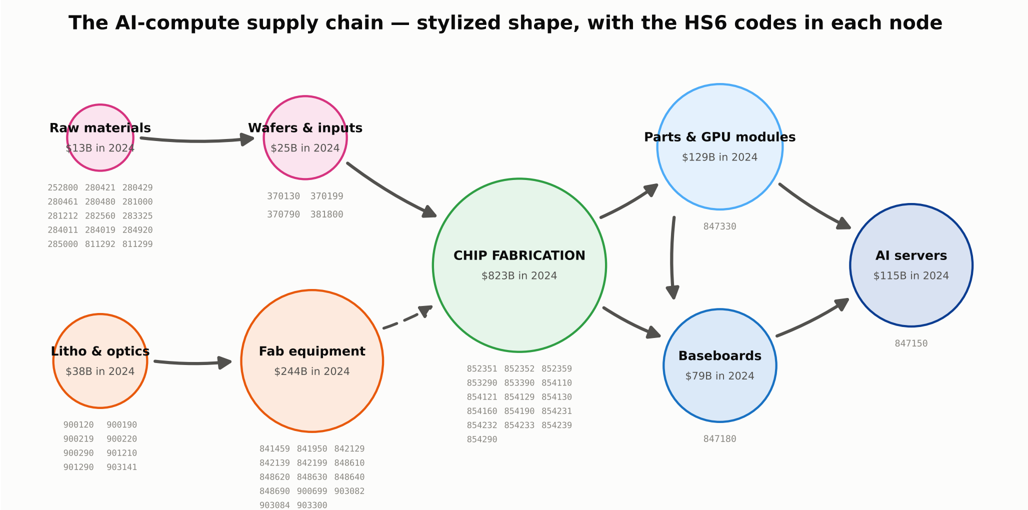
2026-08-20

1. The supply chain in 2024

2024 annual trade flows from the Trade Atlas. We record all relevant HS6 codes, so its coverage extends beyond solely AI compute. China and Hong Kong are counted as one trade bloc (CHK) throughout.

Topography of the supply chain

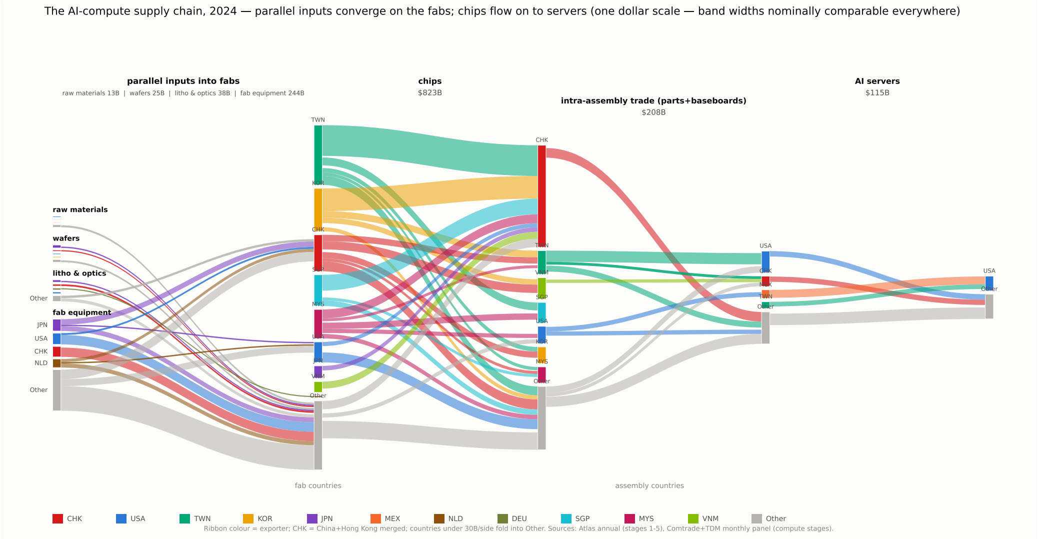
Two upstream inputs feed chip fabrication: capital equipment in orange is used to build manufacturing facilities, raw materials and wafers are consumed per chip. Green represents the actual chip fabrication, which feeds downstream assembly toward AI servers in blue. Circle size in proportion with the log of 2024 USD total trade value.



Two input branches, one output line. Materials (pink) -- silicon, chemicals, wafers -- are used up with every chip made (solid arrows). Equipment (orange) -- lithography and other fab tools -- is investment in capacity: in Taiwan and Korea, equipment imports rise six to twelve months before chip exports (dashed arrow). Korean memory and Taiwanese chips are packaged together, usually in Taiwan. Chip design earns as a service and crosses no border. Circle diameters grow with the log of 2024 world trade. Blue marks the three codes of our monthly panel (the Fed AI-compute basket). A few niche inputs (lasers, vacuum pumps and valves, chip substrates) fall outside the 60 codes. Codes: OECD semiconductor mapping + Fed basket (docs/data.md, section 1).

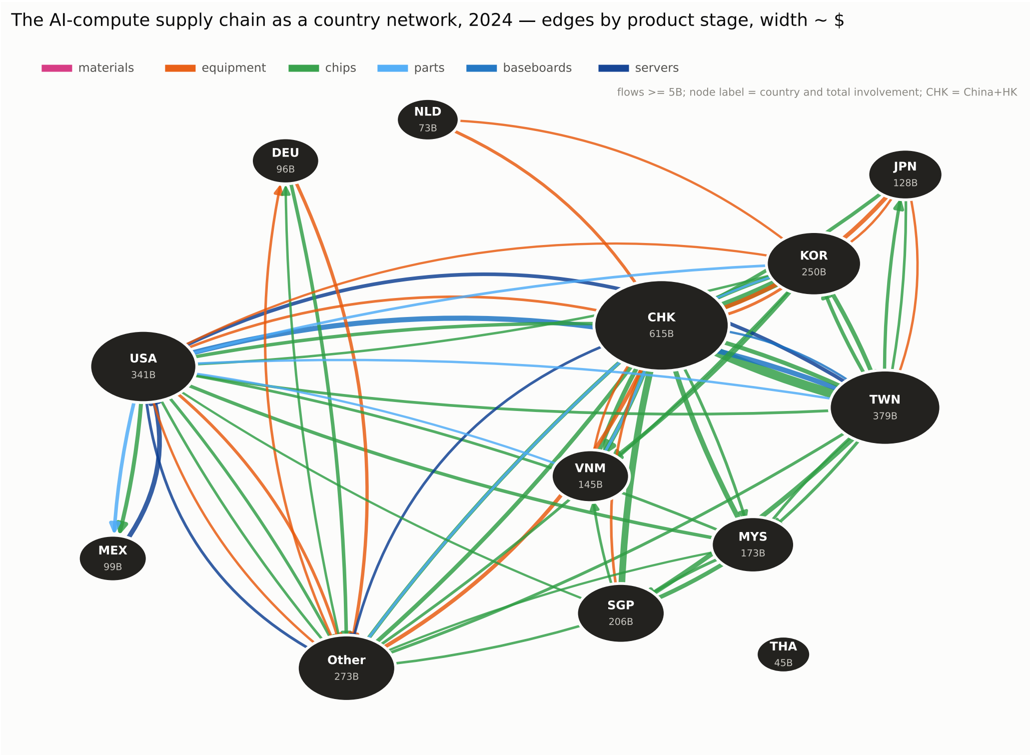
The chain in bilateral trade flows

Width of band = 2024 USD trade. Small flows are grouped into other (grey).



The chain as a graph network

Countries are represented as nodes and directional flows as edges, colored by stage.

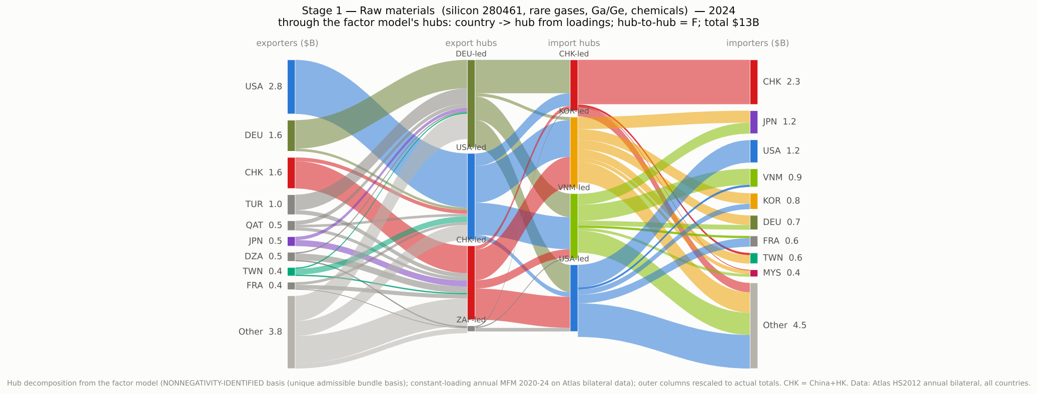


2. The chain in stages

One page per stage: exporters on the left, importers on the right, with the factor model's country groupings in between. The groupings fit the 2024 flows closely (R-squared 0.95 to 0.99). All eight stages are Trade Atlas annual data.

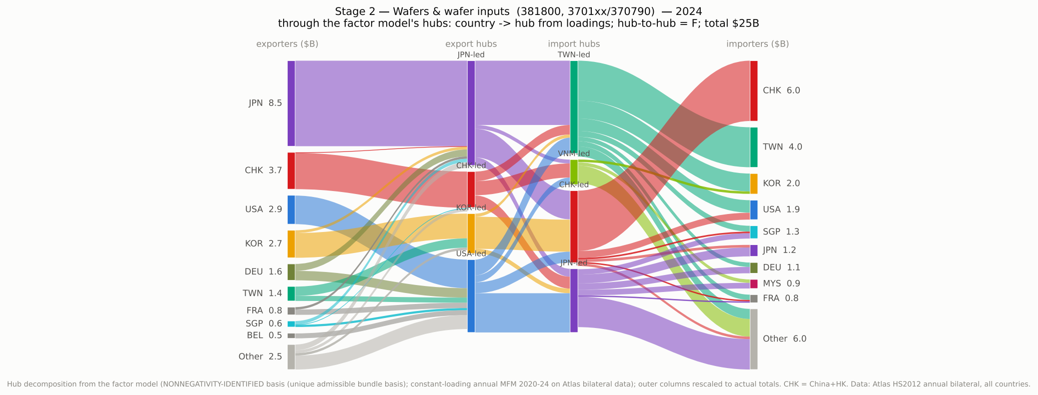
Stage 1 – raw materials

Reading guide for all stage pages: left = exporters, right = importers, middle = the model's export and import groups. Much raw-materials trade is not chip-related (the codes also cover welding gas, abrasives, fertilizer inputs), so flows scatter widely.



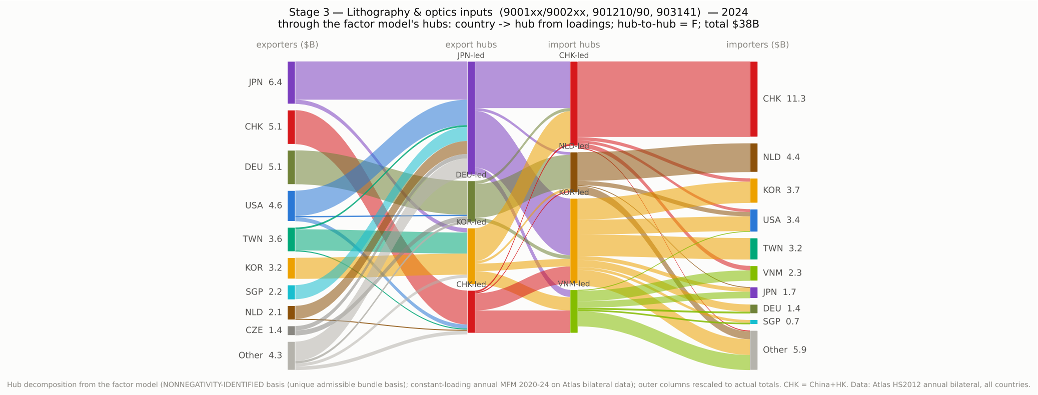
Stage 2 – wafers and wafer inputs

Japan is the largest exporter, about a third of the 25B world total.



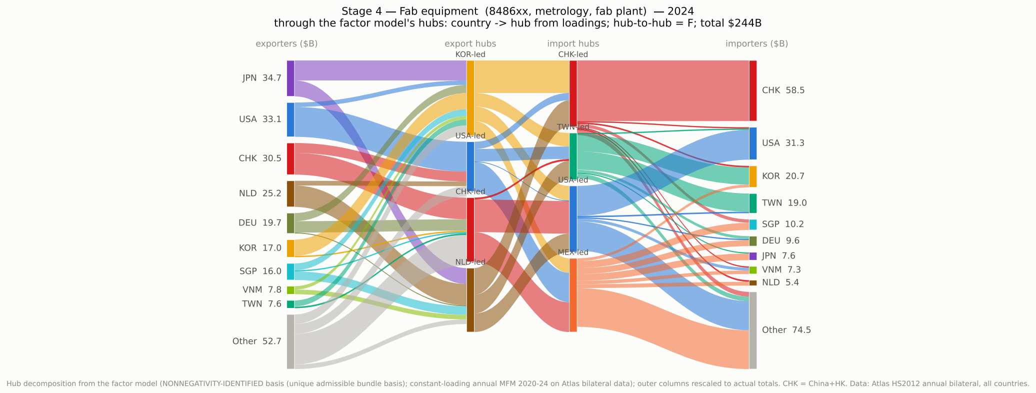
Stage 3 – lithography and optics inputs

Lenses, optical elements, electron microscopes and wafer inspection tools. No dominant supplier: Japan, the China bloc, Germany and the United States each export 5 to 6B.



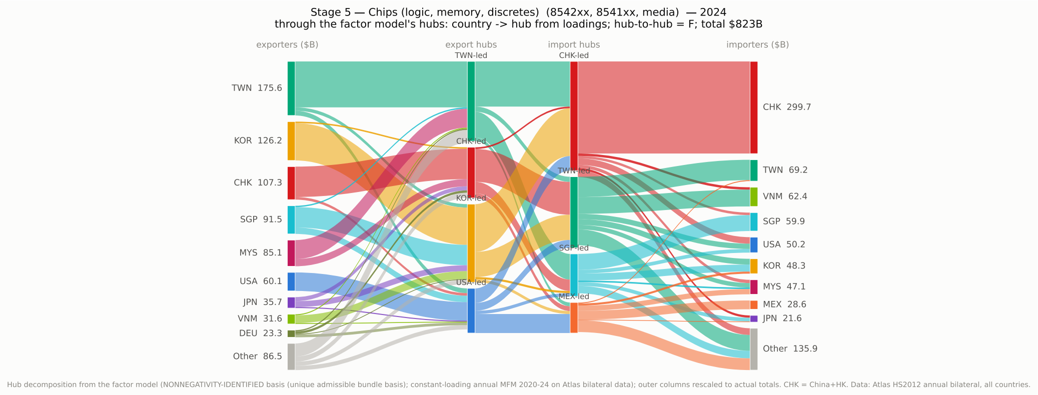
Stage 4 – fab equipment

Machines, not materials: countries buy these to build capacity, and their imports rise 6-12 months before chip exports.



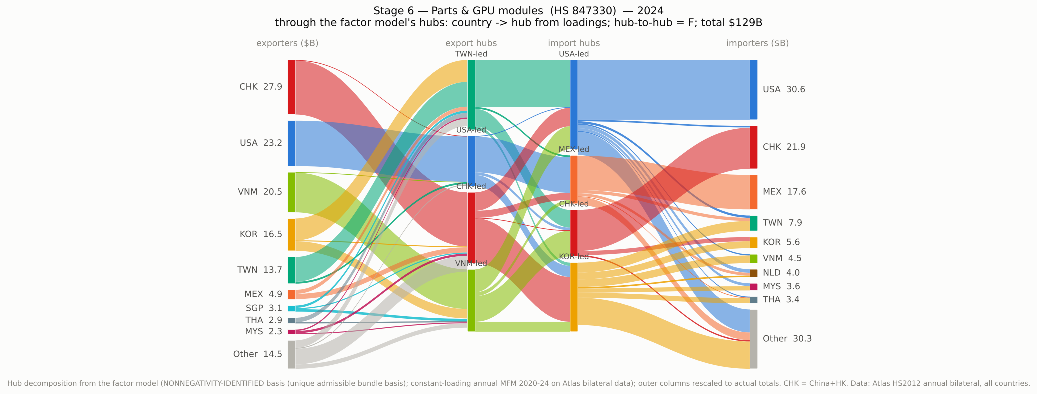
Stage 5 – chips

The biggest stage, and the whole electronics industry – phones, cars and PCs, not just AI. Most chips flow to Asian assembly; the China sphere absorbs about 43%.



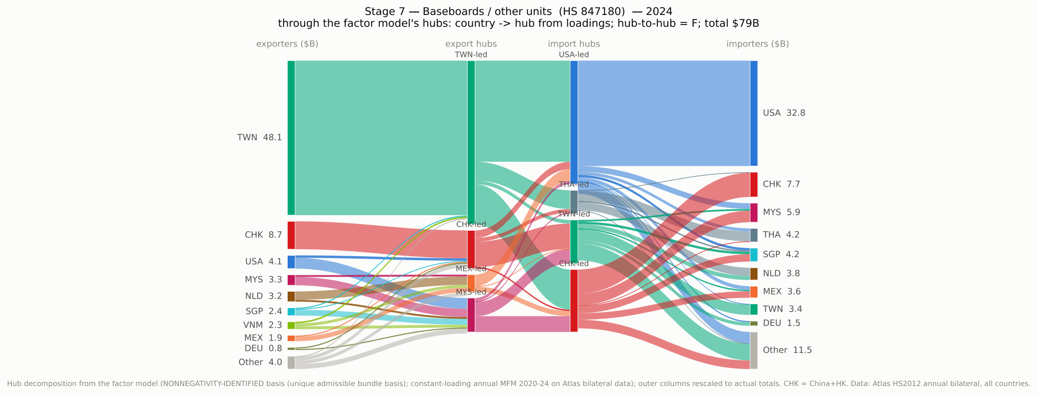
Stage 6 – parts and GPU modules (847330)

The China bloc exports the most by value (28B in 2024), ahead of the United States and Vietnam; the United States is the largest buyer (31B), then the China bloc and Mexico.



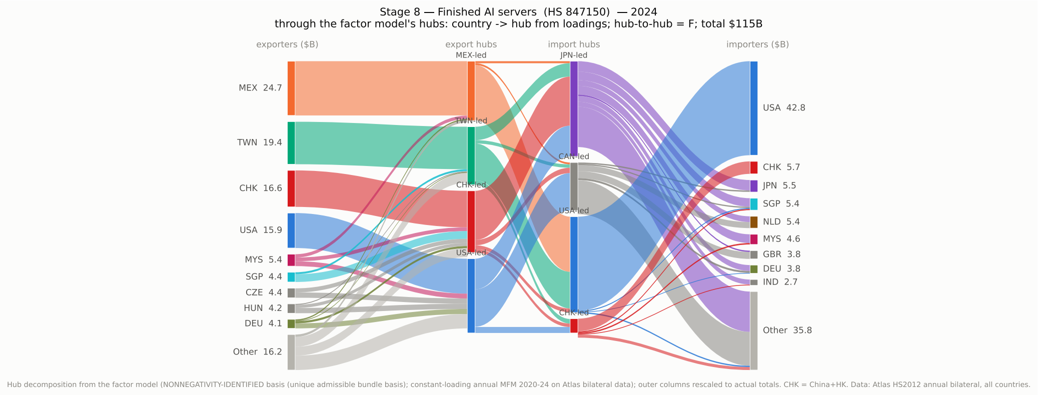
Stage 7 – baseboards (847180)

The most concentrated stage: Taiwan exports 48B of the 79B total in 2024, five times the next exporter, and the United States buys 42 percent of world imports.



Stage 8 – AI servers (847150)

Mexico (25B) and Taiwan (19B) assemble and ship; the United States buys 43B, 37 percent of world imports in this code.

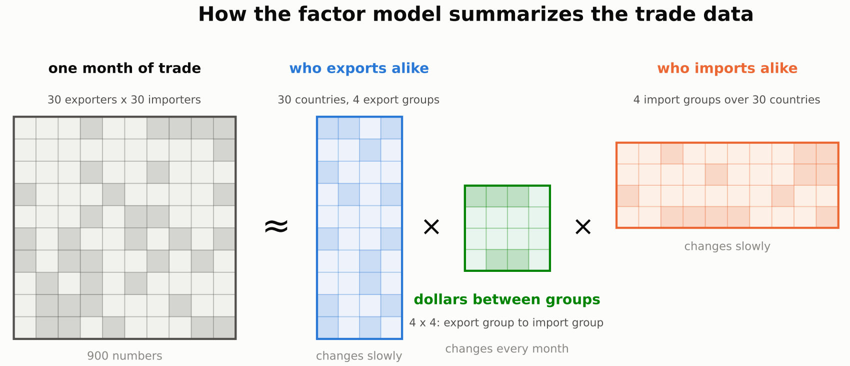


3. The factor model over time – AI-compute codes only

This section uses only the three blue codes (parts/GPU modules, baseboards, AI servers), monthly from 2017. Unless the page says otherwise, China and Hong Kong are counted as one. Shaded bands mark the periods between structural breaks.

How the model works

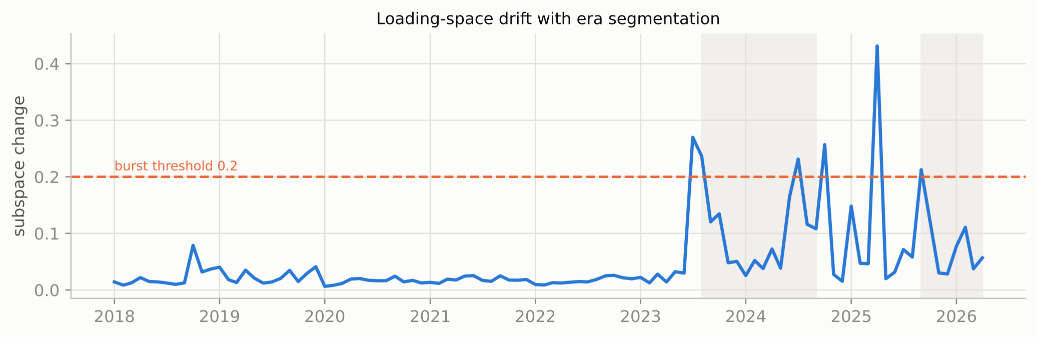
The one idea behind every page that follows.



Every month, 900 country-to-country flows are compressed into: groups of exporters that move together, groups of importers, and a small table of dollars between the groups. The groups are the slow structure of the network -- when they shift, the model marks a break. The small table is the fast part: its cells are the trade channels, and three of them carry the AI boom.

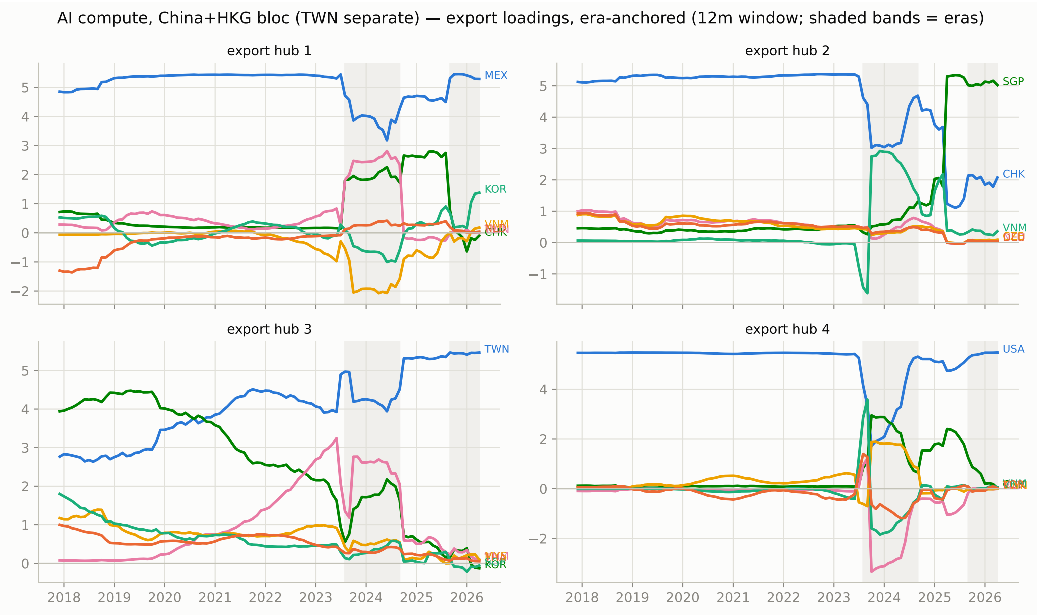
When the network changed shape

The line measures how much the export groupings moved from one month to the next. Flat for six years, then: Aug 2023 (AI demand), Oct 2024 (the quarter memory export controls were signaled), Sep 2025.



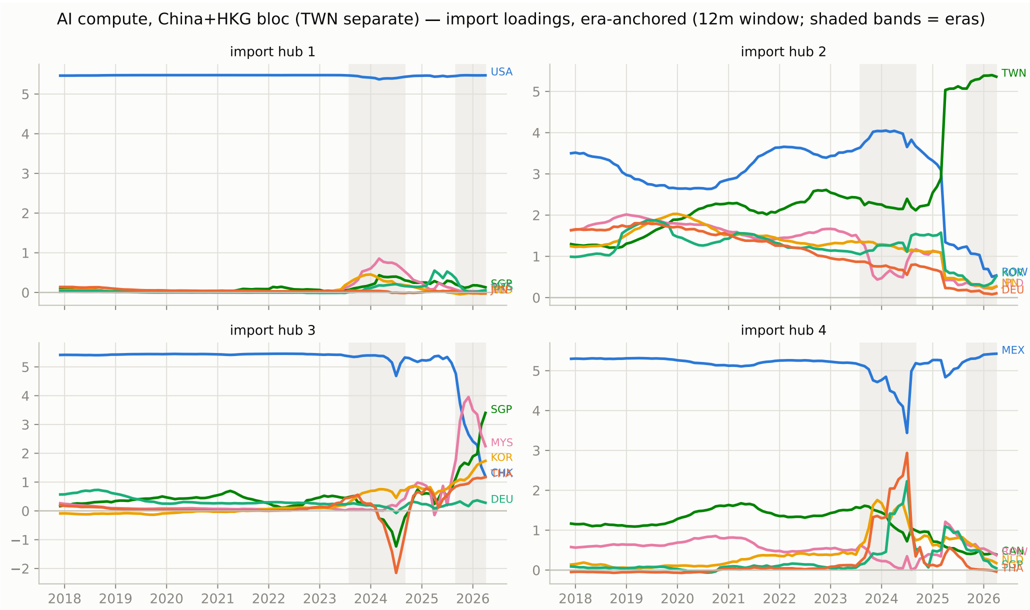
Who is in each export group

At Oct 2024 the China group loses its separate identity and a Singapore-led group appears.



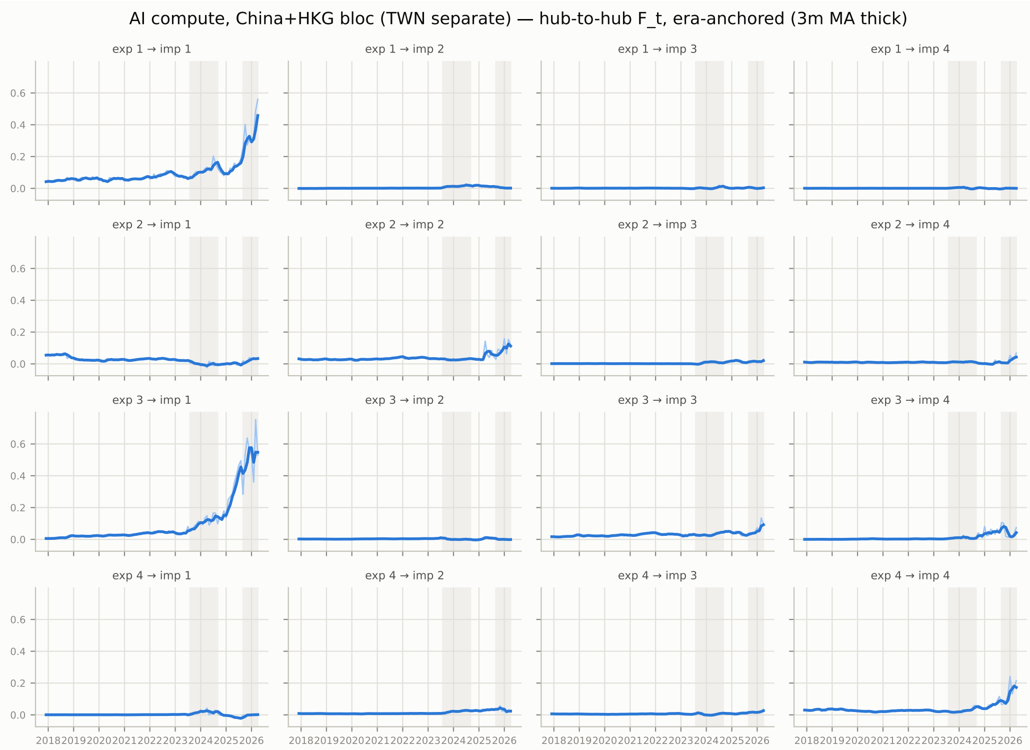
Who is in each import group

The import side is steady: a US group, a Mexico group, a Taiwan group, and the rest of the world.



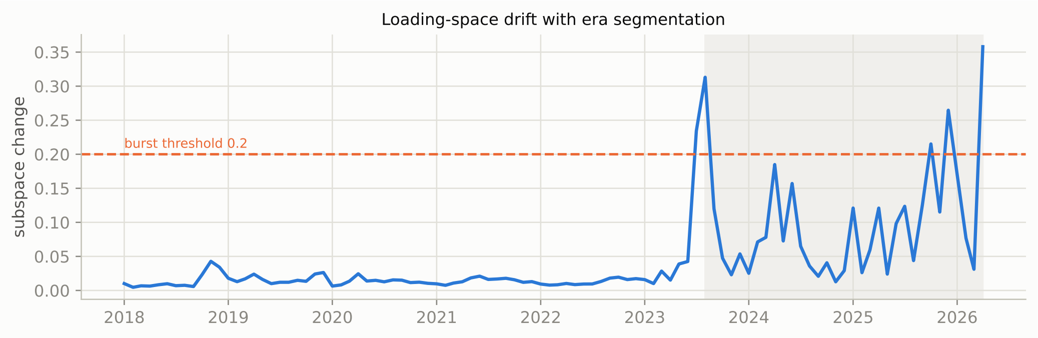
Dollars between groups, month by month

Three of the sixteen channels carry the AI boom: Taiwan to the US, and both legs of the US-Mexico assembly loop. The rest are flat.



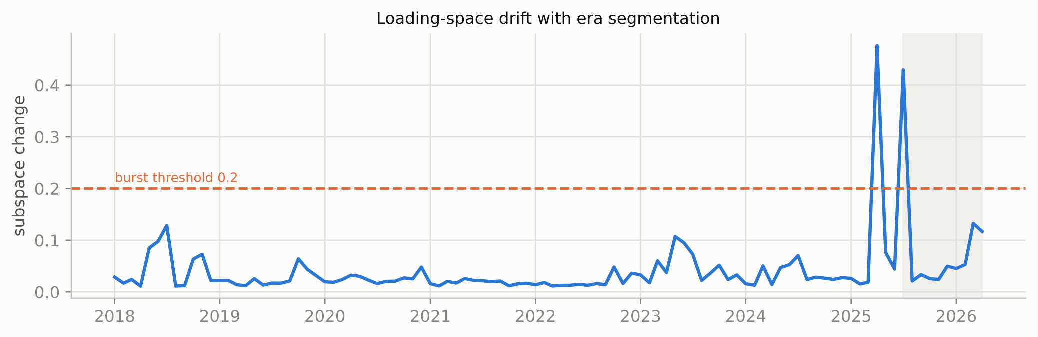
Same, with every country separate

Without merging China and Hong Kong there is one break in the whole sample: Aug 2023. Demand changed the country-level structure; policy did not.



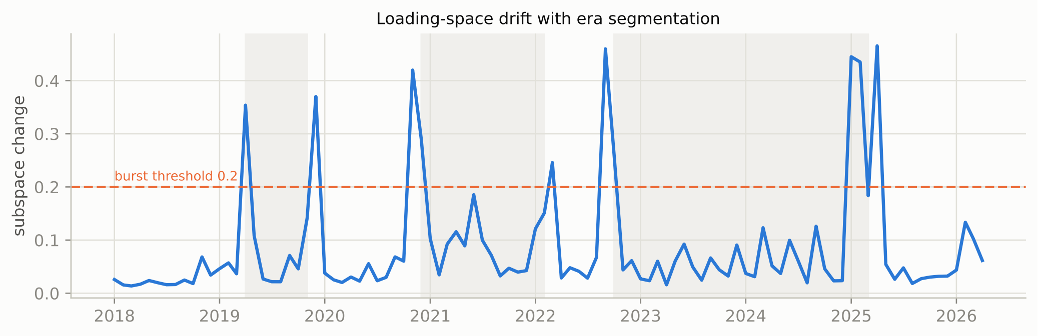
Parts only (847330)

91 months without a break – the stable base layer of the network.



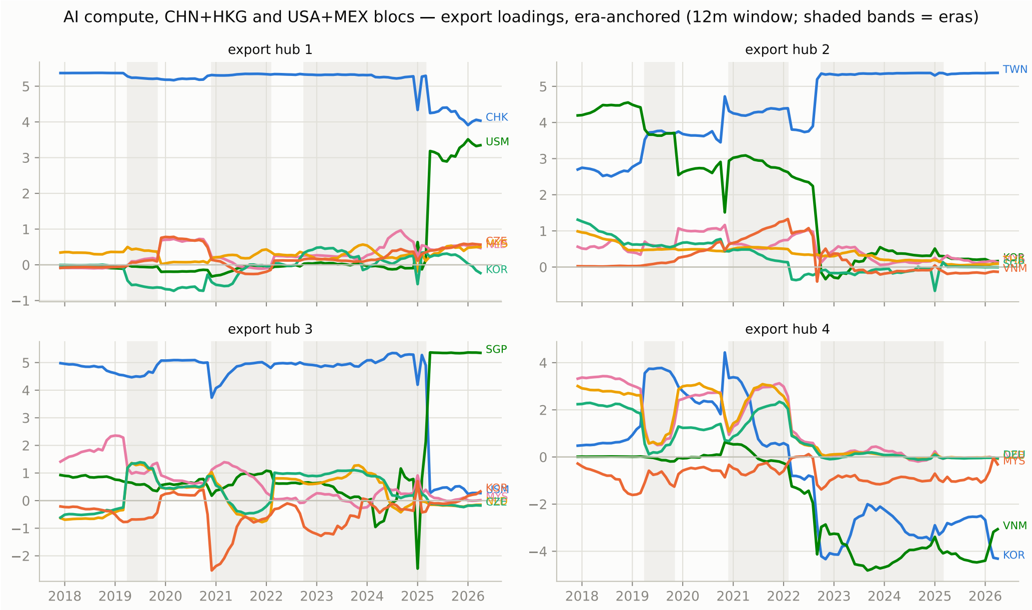
With both blocs merged

Merging China+Hong Kong and US+Mexico hides churn inside each bloc and isolates what happens between them. The breaks line up with policy: the 2019 tariff war, the Oct 2022 export controls, the Apr 2025 tariffs.



The tariff signature

After Apr 2025 the two blocs move together on a single factor – seen at no other break.

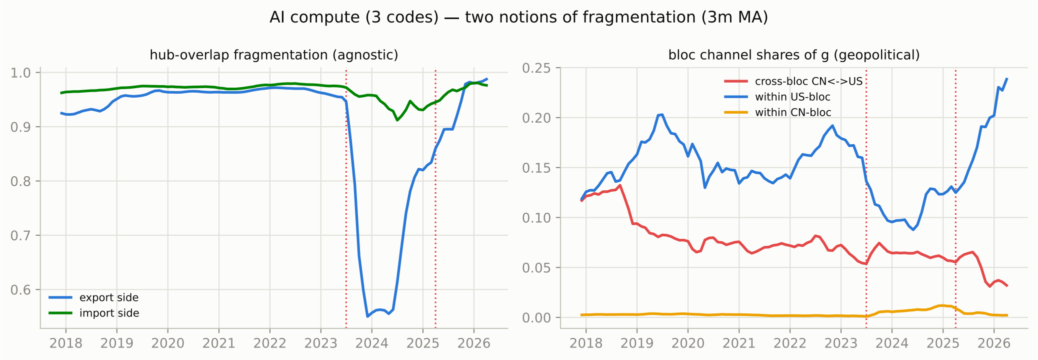


4. Concentration & fragmentation – AI-compute codes only

Monthly measures built on top of the model, for the three blue codes only. Dotted lines mark Jul 2023 and Apr 2025. Interpretation: results/network_stats/findings.md.

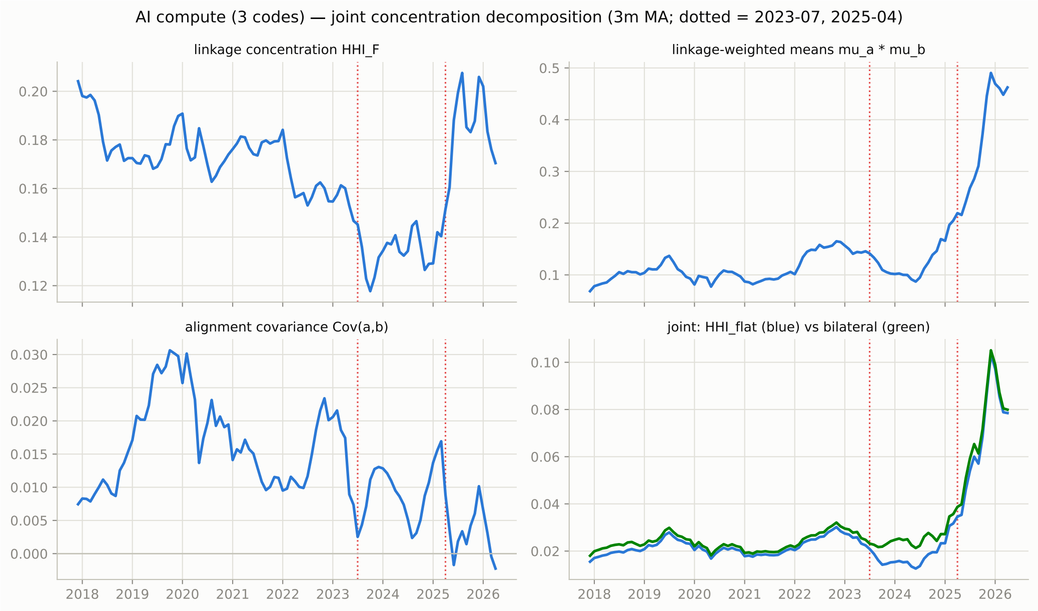
Two meanings of fragmentation

Left: do the model's groups overlap (a technical check, little economic signal). Right: the share of trade crossing between the China and US blocs – down about 70% since 2017, in two steps (2019, then 2023 on), with no recovery between.



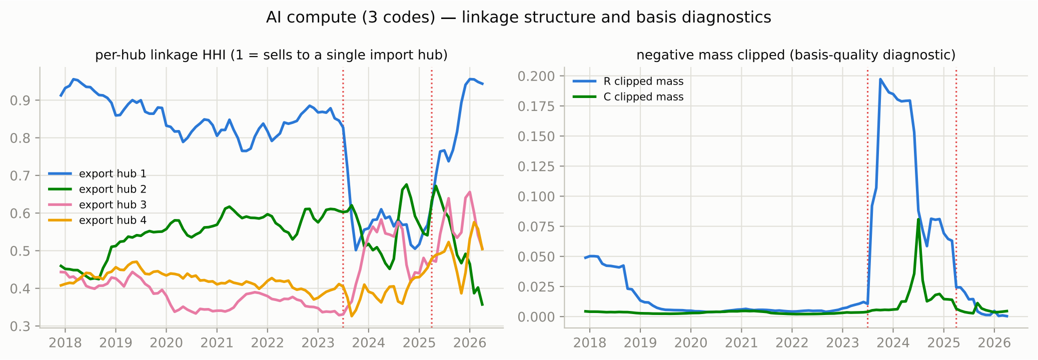
Concentration, taken apart

Overall concentration splits exactly into three parts. The interesting one asks: do concentrated sellers ship to concentrated buyers? Mildly yes, peaking around 2019, fading through the AI era.



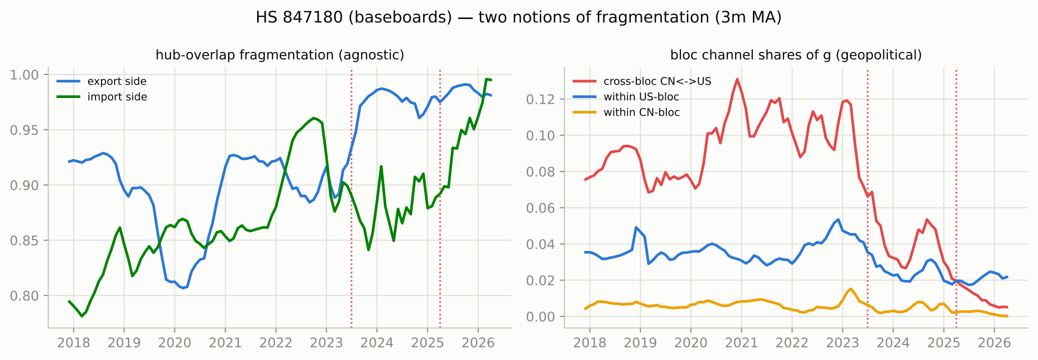
One buyer or many?

Left: does each export group sell into one import group or several. Right: a data-quality check, elevated during the 2023-24 transition.



Baseboards: the sharpest split

The cross-bloc share collapses about 20x from its 2021 peak – the code most exposed to export controls.



Parts: still shared

The cross-bloc share halves but stays the highest of the three codes: the parts trade remains the layer both blocs still share.